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A Comparative Review of Expert Systems, Recommender Systems, and Explainable AI

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Abstract—Previously Expert Systems (ES) dominated Artificial Intelligence (AI) applications and various ES were developed in multiple domains. However, due to knowledge acquisition bottlenecks, these systems fell out of use. With the rise in Machine Learning (ML) and Deep Learning (DL) approaches, another category of systems called Recommender Systems (RS) is now developed for various application domains. As ML/DL systems acted like black boxes, explainable AI (XAI) came into the picture to provide explanations for the recommendations or predictions made. In this paper, we review the architectural similarities and differences between these three approaches along with applications and future directions. It is important to study these to predict the future of RS and any possible resurgence of ES, developments in XAI and application domains.

Index Terms—Expert Systems (ES), Artificial Intelligence (AI), Recommender Systems (RS), Machine Learning (ML), Deep Learning (DL), Search Engine (SE), Rule Engine (RE), and Explainable Artificial Intelligence (XAI).

I. INTRODUCTION

An Expert System (ES) is a piece of computer software created to solve typical problems in a particular field of expertise. ES is a subset of AI. The early ES, which were amongst the first successful practical approaches to AI, were developed in the 1970s. They were applied to tackle difficult challenges of expert decision-making by extracting knowledge from experts and storing it as both facts and heuristics in a knowledge base, in a manner similar to a human expert. ES were expected to capture expert knowledge on a certain topic and are capable of solving a specific set of problems in that domain. These systems were tailored to a particular field, such as medicine or science, accountancy, agriculture, bio-engineering, Geology, Finance, etc. These are some of the disciplines where ESs have been developed [15] [20] [19]. However, Knowledge Acquisition remains a bottleneck for ES. From the early years

of the new millennium, Machine Learning (ML) methods have become more important. Recently, Deep Learning (DL) has gathered momentum and several applications are now being developed. While these three approaches look different, there are certain similarities that we will like to bring out in this paper.

This paper is organized as follows: Section-II introduces Expert Systems whereas section-III describes the architecture of the Expert Systems and explains the relationship between ES and RS. Section-IV provides the overview of Recommender Systems and Section-V provides the overview of Explainable Artificial Intelligence. Table-II describes the various applications of Recommender Systems, Expert systems, and XAI.

II. EXPERT SYSTEMS

An ES is computer software designed to tackle complicated problems in a given domain that are typically solved by human experts. An ES was modeled to be like a human brain, simulating the thinking of a human expert. An ES was to undertake analysis, design, monitoring, and decision-making, among other things. For example, when we access a website for searching items or products using keywords with typographical errors, ES recommends the user to correct those keywords for searching. Nowadays Google, Amazon, Flipkart, banking, and other E-commerce companies build their own systems like RS or ES for the convenience of their customers.

Expert systems were said to be quite mature and thus successful [22]. ES were expected to outperform skilled humans in medical diagnosis, predicting crop disease, and extracting minerals from ore. DENDRAL is the first ES to investigate how scientists generate hypotheses and uncover new ones [50] and it was used to determine the structure of chemical compounds. Later many ES were designed in like, MYCIN,

TABLE I
DIFFERENT TYPES OF EXPERT SYSTEMS AND THEIR APPLICATION AREAS

Expert Systems	Developed	Application Area
DENDRAL	J.Lederberg, Feigenbaum, 1965	Chemistry
MYCIN	Edward Shortliffe,1976	Medicine
ACE	Vesonder,1983	Electronic
AIRPLAN	Masui,1983	Military Science
PROJCON	Underwood,1981	Information Management
IMACS	O'Connor,1984	Manufacturing
LDS	Waterman, 1980	Law
SPERIL-II	Ogawa,1984	Engineering
DART	Lars BaK,2011	Computer Systems

EMYCIN, THEIRESIUS, GUIDON, PUFF, PROSPECTOR, PXDES, and CaDeT [50].

With the rapid advancement of technology, researchers are now combining a variety of approaches such as AI, NLP, ML, and DL to construct ES that can interpret textual content. Unfortunately due to excessive hype and other reasons ES did not live up to corporate world expectations during the early days of AI. However, with the rapid advancement of AI, ML, NLP, DL, and ES might come back. They may go by other names now, but the principle of tackling expert-level issues remains the same.

A. Definition and Description of ES:

The British Computer Society's specialist group proposed the formal definition of ES:

Definition: "An expert system is defined as the incorporation of a knowledge-based component derived from an expert skill into a computer in such a way that the system may provide intelligent advice or make intelligent decisions regarding a processing function. The on-demand capability of the system to justify its line of reasoning in a manner directly understandable to the enquirer is a desirable additional attribute that many would consider basic. Rule-Based programming was used to achieve these properties [22]."

Expert systems are also called as 'Rule-Based Systems'. In these systems knowledge is converted into production rules, i.e **IF THEN** is a rule format. The Rules can be described with the left-hand side containing antecedent or condition part. Right-hand side contains action part or conclusion part. It is shown in the below format [50].

Let *C*-indicates conditions (*C*1,*C*2,*C*3...), *A*-indicates actions (*A*1, *A*2...).

IF: *C*1 and *C*2 and *C*3

THEN: Take *A*1

For example

IF: The triangle has equal sides and the internal angle is 60 degrees.

THEN: It is called an equilateral triangle.

and here each rule describes the specific domain knowledge regarding human expertise. It stores the number of inference rules in the form of known 'facts' and useful 'conclusions'. According to this if known facts support the condition rule, then the action part or conclusion rule is

accepted. IF/THEN/ELSE structures also exist in high-level languages, those who are familiar with procedural computer languages like FORTRAN, C, BASIC, and PASCAL are familiar with IF/THEN/ ELSE algorithms. Fig 1 describes

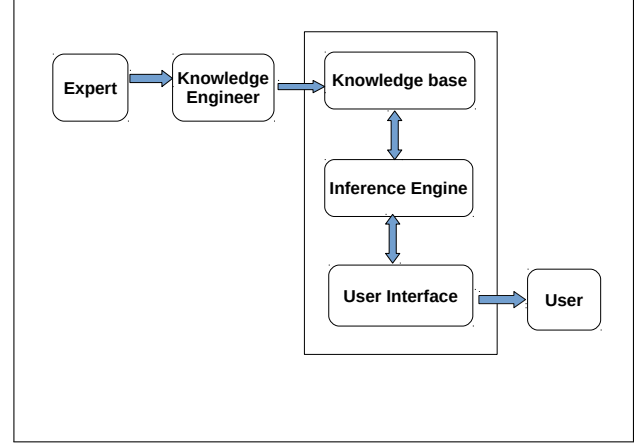


Fig. 1. General frame work of the ES [1]

the general framework of an Expert System. Here human experts may be anyone like a doctor or a domain specialist. For example, a patient consults a doctor and is examined for the symptoms of the patient and is prescribed medicines. In expert systems, human experts examine the patient and diagnose diseases and propose solutions to the knowledge engineer and the knowledge engineer puts the information into the expert modules to develop the expert systems. The following describes the main modules of the Expert Systems.

• Types of ES:

There are five different types of expert systems. These types include *Rule-based ES*, *Frame-based ES*, *Fuzzy-based ES*, *Neural ES*, and *Neuro-fuzzy ES*.

B. Modules of Expert Systems (ES)

The three main modules of Expert Systems, can be explained below [1].

- **Knowledge Base (KB):** Here KB is the main module of an ES and it stores the specific knowledge of one domain. An ES's knowledge base incorporates both factual and heuristic information.
- **Inference Engine (IE):** It is a tool that allows one to reason about data in a knowledge base. Its purpose is to make a recommendation, and it accomplishes this by combining the facts of a specific case (input data) with the knowledge in the knowledge base. Semantic networks, production rules, and logic statements can all be used to make inferences. Forward and backward chaining are two forms of data-driven methods. Forward chaining is used to generate predictions, whereas reverse chaining is used to figure out why a specific action occurred.
- **User Interface (UI):** Interaction between the program and the users is provided by the hardware and software of the ES.

Here we describe a detailed explanation of the architecture of ES in Section-III.

III. ARCHITECTURE OF EXPERT SYSTEMS (RULE BASED LEARNING (RBL))

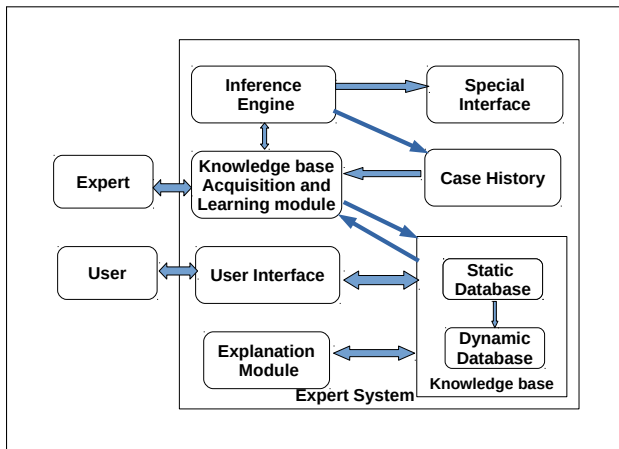


Fig. 2. The general architecture of expert systems [50]

In Fig 2, the general ES architecture is given. Below we discuss about various components present in the architecture.

- **The term Inference Engine:** It is a word that refers to the process of searching through a knowledge base and inferring new knowledge using various algorithms. The inference engine uses the following methodologies:
- **Forward Chaining:** It starts with known facts before asserting new ones. Example: PROSPECTOR ES used for identifying the molecular structure of unknown soil or ore using forward chaining strategy.
- **Backward Chaining:** It starts with the end in mind and works backward to see what facts must be presented to meet the objectives. Example: MYCIN ES used for identifying infectious blood diseases using backward chaining strategy.
- **Knowledge Acquisition and Learning Module:** The knowledge present in the expert system may be obtained from various sources such as textbooks, human experts, reports, presentations, videos, internet, etc. This module allows the system to acquire more knowledge regarding the particular problem and store it in the knowledge base.
- **The term User Interface:** An expert system's user interface allows users to communicate with the system through an interface, which aids in the system's creation of knowledge for the task at hand and its querying.
- **Explanation Module (EM):** It allows the user to ask the expert system questions about how it came up with the answer and why.
- **Special Interface (SI):** This module is used for performing an activity such as handling uncertain knowledge. It handles uncertain data and uncertain knowledge.
- **Case History (CH):** It stores files created by an inference engine using a dynamic database and it is used in a learning module to enrich its knowledge base.

- **Knowledge Base (KB):** KB of an expert system consists of knowledge regarding the problem. The memory is in the form of a static and dynamic database. Static knowledge consists of rules and facts. Dynamic knowledge consists of detailed information where many questions are asked to the human expert related to the problem. In practice, all the accurate solutions are put in the dynamic database.

Advantages:

Unlike human experts ES must be repeatable, and they react quickly to specific queries. The performance of an ES system is unaffected by emotions, tension, or exhaustion. When the Knowledge base contains accurate information, the likelihood of making a mistake is lowered. In difficult situations when human presence is not safe, ES can be used.

Disadvantages:

It's difficult to explain the logic and reasoning behind the decisions in ES. When it comes to making decisions, ES does not use common sense. Human specialists' ability to respond creatively is lacking. Complex tasks are difficult to automate with ES. ES is unable to identify the absence of a response. ES is not versatile and flexible enough to change with the environment.

A. Reasons for Expert Systems Being Discontinued

An ES is very hard to develop for the researchers. Once the ES is built it cannot be modified easily. It is very expensive to build and it's a time-consuming process. Thus ES fails due to competitive pressure, enthusiasm, lack of momentum, wrong analysis, and misunderstanding of the problems [11]. One more reason that ES failed is due to the developers did not showing interest to train staff to maintain ES. During the development phase, the developer did not show sustained interest to train its employees in the IT field. The following, passage describes the relationship between ES and RS.

B. The Relation between Expert system and Recommender System

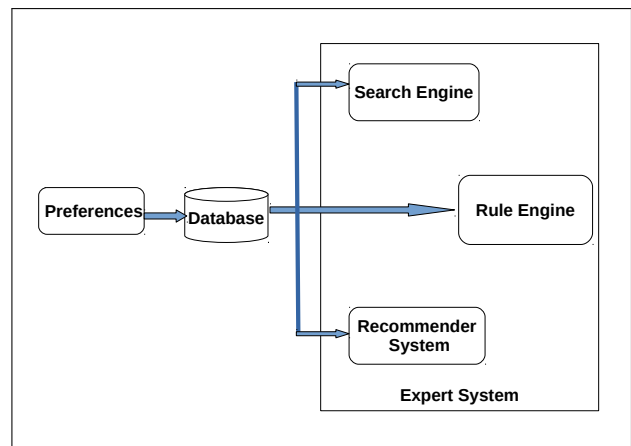


Fig. 3. The Relation between Expert System and Recommender Systems

In Fig 3 we describe the relationship between ES and RS. The user preference about the data/items from the database in an expert system suggests the corresponding data to the user. It has these components.

- **Search Engine (SE):** Search engines provide useful and precise information in response to user inputs.
- **Recommender Systems (RS):** Recommender Systems provide suggestions to the users, tailored to their needs.
- **Rule Engine (RE):** It is used to understand procedures and their rules.

In this paper section-IV describes the overview of Recommender Systems.

IV. RECOMMENDER SYSTEMS

A recommender system (RS) is an information filtering system that tries to predict a user's "rating" or "preference" for a given item. The domain and specific properties of the data dictate the architecture of these recommendation engines. A Recommender System (RS) is a tool that can help to narrow down a users alternatives from a large list of possibilities [63].

A. Definition and Description of RS:

The goal of a recommendation system (RS) is to give useful suggestions to a group of users about items or products that they might be interested in. Beyond the RS, the goal is to establish a relationship between products and consumers, as well as to make decisions about which product or items are best for each user [40]. In the real world, application of industry strength RS is for books on Amazon, or movies on Netflix [39], etc.

In Fig 4 we describe the general framework of RS. Here,

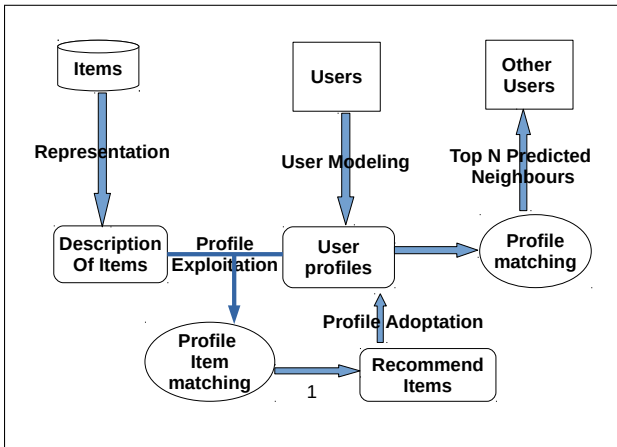


Fig. 4. General frame work of RS [40]. The marker '1' indicates the top N predicted items.

RS gathers information about the user utilizing a variety of ways and sources to predict the user's needs and make recommendations based on the analysis [53]. In Fig 4 We give the general notion of an RS while comparing the user profile to that of other users, calculating the common items between them, and recommending the differences [33]. Recommendation engine software employs eight ways to predict

what a user would want to buy, listen to, or watch. These eight strategies are Content, Collaboration, Demographic, Hybrid, Knowledge-based, Context-aware, Semantic-based, and Session-based Recommender Systems.

In our research, we find that RS and its applications in various domains have quietly become very prominent in many human-centric applications such as 'Assistant to a Travel Guide', 'Similarity in Math Equations', and 'Gamification and IT Addiction'. Many types of ML and DL models are used for choosing the products/items. RS suggests the list of similar or dissimilar products, rating of the products, and experience of the product for the new users. RS internally uses ML or DL approaches to build the matches for queries for the recommendation. There are three phases in RS: The information collection phase, the Learning phase, and Predicting/Recommendation phase.

• Types of RS:

There are eight different varieties of RSs. (i) Content-based RS, (ii) Collaborative-based RS, (iii) Demographic-based RS, (iv) Hybrid-based RS, (v) Knowledge-based RS, (vi) Context-aware RS, (vii) Utility-based RS, (viii) Session-based RS's, Here we give few examples.

• Assistant to a Travel Guide:

It aids humans in locating external objects as well as the ability to construct routes to these objects. In terms of multimedia features, electronic maps, GPS navigation, and tracking are used for the UI. It's aim is to give tourists the finest way possible and recommendation of restaurants, hotels, and planning [8].

• Similarity in Math Equations:

Similarity of mathematical equations, structural similarities, patterns, approaches, and mathematical equations similarities evaluations were seen [54].

• Gamification and IT Addiction:

A condition known as IT-Addiction was seen by using the example of watching TV episodes available through online video streaming services [9]. To improve and simplify access to information services, Gamification [14] is one of the most widely studied topics today, with a significant transdisciplinary component. "Gamification has been utilized in a variety of research fields and areas to improve on traditional approaches." According to [47], which includes software development, engineering, and software testing. [17] [52], e-learning [49] and social adaption [9].

Advantages:

RS provides easy recommendations to users for quickly searching items to save their time. Most of the time RS provides the user reviews giving accurate information regarding the items. RS does not use user data for recommending items. Similar things are matched between users and the system. RS are also used commercially to increase the customer satisfaction rate [40].

Disadvantages:

If the RS makes biased product recommendations, the

consumer will not be able to obtain the correct items. On some websites, RS are providing suggestions that are not relevant to the items based on analysis of less information. The quality of products depends on more ratings [40].

In recent times even RS seems inadequate and have not been applied in general to domains where explanations or justifications for AI Systems or the conclusions or choice is needed. The following Section-V describes XAI.

V. EXPLAINABLE AI (XAI)

Due to the huge availability of data sets and more computing power ML systems are now performing well. AI system performance in a wide range of tasks is rapidly improving, as seen by advances in picture recognition, speech analysis, strategic game design, and other areas. The issue with modern models is a lack of interpreting ability, trust, and transparency. These are the main drawbacks of several domains such as health care, finance, consultancy, and many more. To overcome these issues XAI has emerged. It is one of the evolutionary development of Information Systems [16].

XAI aims to improve the user experience of a product or service by assisting end-users in having faith in the AI's decision-making abilities. The XAI system explains its understanding and capabilities. It explains what it has done, what it is doing now, and what will happen next, as well as revealing the key facts on which it is acting [4].

A. Definition and description of XAI

D. Gunning defines XAI as "A set of machine learning techniques that help humans understand, trust, and manage the new generation of artificially intelligent partners [30]." The similarities, ideas, and differences between the terminologies used in AI and XAI are discussed in this section.

- **Understandability:** The word refers to the model's qualities and purpose in helping humans understand it. The model's operation, based on which algorithms we implement, and the internal structure of the model, these parameters are to be considered for understanding the model [41].
- **Comprehensibility:** When building ML/DL models it refers to a learning algorithm's ability to learn, based on their knowledge in the human-understandable manner [13], [21], [25]. Here XAI works based on three principles: Explainability, Transparency, and Interpretability.

In the following, we describe these principles.

- **Explainability:** According to [27], an explanation is a human-comprehensible interaction between a decision-maker and others that is both accurate and comprehensible. It is the process of gathering features from the interpretable domain that have contributed to the production of a decision in a given example (eg. classification and regression). In AI, there are two kinds of machine learning algorithms: white box and black box ML methods. The white box,

machine learning algorithms produce findings that are understandable to domain experts. ML algorithms in the black-box form are difficult to describe and comprehend for domain specialists.

- **Transparency:** The designer can define and justify the technique for extracting model parameters from training data and producing labels from testing data. Three types of transparent models are available. Models that can be simulated, that are decomposable, and those that are algorithmically transparent [36].
- **Interpretability:** It is to describe the capability of comprehending the machine learning model and to present the underlying reason for decision making in a human-friendly manner [24].
- **Data Security (DS):** The General Data Security Regulation of the European Union has an "right to explain" clause that will demand the use of XAI [26].

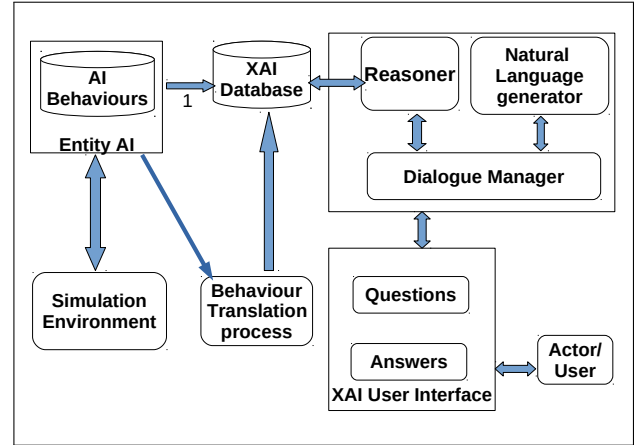


Fig. 5. XAI architecture [12]. The marker '1' indicates the scenarios and logs.

In Fig 5 We describe how an XAI system is configured to allow users to choose a piece of matter which speaks to communicate within a time point. Users then select questions from a menu to ask the entity about the current time point. The conversation (dialogue) manager directs the system's answer, first retrieving relevant facts with the reasoner and then providing English responses with the natural language generator (NLG). NLG populates natural language templates with data from databases and human interaction [12]. The algorithms utilized in XAI are divided into two categories: white box and black box ML algorithms.

Advantages:

XAI could be a social right to explanation implementation. Even though there is no legal right or regulatory obligation, it is important [26].

- **Improved Transparency and Explainability:** XAI used in the business world advanced AI models to perform better and to be aware of why they behave in a

certain manner for specific conditions. If it is a black-box model, the human can use an explanation to understand how these AI models achieve a certain conclusion [2].

- **Faster Adaption:**

XAI used in the business sector aims to understand the AI models better ways, and trust them to make the most important decisions [2].

- **Improved Debugging:**

When suddenly a system works or suddenly the system stops working, XAI can be used to detect the problem and help the developer to resolve the issue [2].

Disadvantages:

In machine learning, there is a black box where even the designers are unable to explain why an AI made a particular decision [2].

Table II describes the various applications domains of Expert Systems, Recommender Systems, and XAI.

The following subsection describes few future directions of XAI which helps the researchers for doing further research.

B. Future directions of XAI

Our intelligent systems will become even more intelligent as AI advances. For example, Tesla cars will be able to drive themselves, houses will be able to adapt their electricity usage, and robots will be able to diagnose human sicknesses, to name a few examples. A few XAI future directions that are currently either limited in functionality or limited scope (research projects) are listed.

- **Assistants with Artificial Intelligence (AAI):**

An AI assistant is a computer program that can recognize natural language voice instructions and perform tasks for the user. Here XAI methods are used for speech recognition and keyword classification [62]: Google Assistant, Siri, Cortana, Amazon Alexa, Bixby, DataBot, Lyra, are just a few examples.

- **Payments that are made Independently:**

Payments in the future and banks, merchants, and payment suppliers must get ready now for independently made payments. Here XAI methods are used to detect anti-money laundering in banking and finance sectors [35].

- **Autonomous Vehicles (AV):**

Without the need for human intervention, an autonomous vehicle can operate and carry out necessary tasks. To properly employ the system's decision-making paradigm in future missions. XAI methods are used to give users explanations that allow them to comprehend the system's general strengths and weaknesses, provide knowledge of how it will behave in future or different scenarios, and maybe allow users to fix the system's errors [29].

- **Autonomous Drones (AD):**

Autonomous drones also called Unmanned Vehicles (UAV) are used for transporting services. Here XAI based simulation system can be built for transporting packages (passengers) and operating in critical scenarios [43].

- **Bionic Organs:**

Bionic organs are artificial organ devices or tools. They will replace damaged natural organs. Here XAI methods are used in the manufacturing of an artificial organ using 3D printing technology [59].

- **Conversational Agents (CA):**

Conversational agents or virtual agents are a type of dialogue system that incorporates Natural language processing (NLP) used for inquiry and replies given in human language. Here XAI is used in designing speech recognition systems for keyword classification to improve human interaction [62].

- **Smart City (SC):**

The Internet of Things is at the heart of the smart city concept (IoT). Constant population growth and urbanization have increased the demand for creative urban planning concepts that have the least possible impact on the environment, and citizen lifestyle. Here XAI methods are used for developing cybersecurity [31], and service exchanges applications [18].

- **Smart Dust (SD):**

Smart dust is a collection of micro-mechanical systems that can sense light, temperature, vibration, magnetism, chemicals, and other stimuli, such as sensors, robots, and other devices. Here is with the help of smart dust sensors. XAI is used in military applications for early detection of chemical and bioagents in war [29].

VI. DISCUSSIONS AND CONCLUSIONS

Before we conclude it is apparent that are several methodological improvements are required for the various components.

A. Methodology Limitations Common to RS and XAI problems with DL

Compared to other techniques DL requires a huge amount of data. It is expensive to train complex data models. It requires GPUs and expensive machines. There is no specific theory for choosing DL tools, or a data model, or a training method, and specific parameters. It is difficult to acquire skilled persons in organizations. It is difficult to identify the output based on learning. It is not possible to solve all problems using convolution neural networks algorithms.

B. Concluding Remarks:

In this paper, we reviewed the architectural similarities, differences between ES, RS, and XAI and their applications. A few domains of RS and aspects of their methodology were discussed. Finally, ES, RS, and XAI were compared. In the future, we expect that RS with better NLP interfaces may perform like ES with XAI.

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TABLE II
COMPARISONS OF ES,RS,XAI AND THEIR APPLICATION AREA

Application Area	Expert Systems	Recommender Systems	XAI
E-Commerce	ES is used for customer purchasing behavior, which is used to detect potential customers' near-future interest in items [7].	Recommendation of items to consumers [57]. Recommendation of movies and videos to customers [28].	Its goal is to influence customer behavior to favor certain items and brands [32].
Transportation	It is to acquire knowledge on how to eliminate automatic and telemechanic system malfunctions in railway transportation [58].	Route recommendation for goods and passengers [23]. It recommend for tourist travel [42].	The constraints on the effective provision of urban roads and the resulting consequences for urban development [10].
Health care	To improve self-care and clinical management, ES will create automated alerts and patient instructions based on telemonitoring data [56].	Recommendation of treatment plan or medical advise [44]. Recommendation of health in mobile systems [48].	Used to analyze and diagnose from health data by AI-based systems. To achieving accountability and transparency [51].
Agriculture	It recommends diagnosing and managing pests for several crops. Taking good decision for farmers [55].	Recommendation of fertilizers to farmers [3]. Recommendation of agriculture product [5].	Used to develop a plant disease detection system [34].
Defence	In this sector, ES do not perform beyond the prototype stage well [61]. So only simple ES is used.	Recommendation to detecting attacks and increase the defense ability [6].	Lethal autonomous weapon systems (LAWS) can do less harm, XAI is becoming more essential in military activities. During warfare, both civilians and combatants are present [60].
Banking	Credit rating weights are applied for each component that determines the credit decision, and ES is used to analyze and support credit judgments in the banking sector [38].	It advises bank clients on relevant products such as credit cards, debit cards, deposits, loans, pension funds, mutual funds, and bank working hours [37].	To detect fraudulent and suspicious activity (consumer credit, credit risk, and anti-money laundering)in corresponding banking transactions [35].
Finance	It is utilized in international commercial transactions to provide currency exchange advice. In both developed and developing countries [46].	RS is utilized to help financial advisors increase their ability to offer a wide range of customized investment and credit options [45].	Used to detect the preliminary investigation in consumer credit, credit risk, and anti-money laundering in the financial sector [35].

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